Final Year Project Stress Chatbot

Educational Support, Not Therapy

This project presents a purpose-built chatbot designed to support final-year students through academic stress — offering instant, safe, and educationally grounded guidance. Built with multiple response strategies and a robust safety filter, it bridges the gap between student anxiety and accessible academic support.

The Problem: Students Are Stressed and Unsupported

Final-year students face a unique convergence of pressures — looming deadlines, supervisor expectations, motivation dips, and project uncertainty. When stress peaks, instant guidance is rarely available, and anxiety compounds quickly.

Deadline Pressure

Students struggle to manage time and prioritize tasks under tight project timelines.

Supervisor Stress

Unclear feedback and high expectations create anxiety and self-doubt.

Motivation Gaps

Without encouragement or structure, students lose momentum mid-project.

No Safe Guidance

Existing tools either offer therapy (unsafe for academic use) or generic, unhelpful answers.

Project Objectives

This chatbot is built with four clear goals — each designed to address a specific gap in student support during final-year projects. The system balances responsiveness with responsibility, ensuring every interaction is both helpful and safe.

1

Instant Educational Guidance

Provide students with immediate, contextually relevant answers to common project-related stress questions — available 24/7.

2

Safety Filtering

Detect and intercept sensitive or distress-related queries, redirecting students to safe, educational content rather than therapy-like advice.

3

Motivation & Productivity

Offer structured encouragement and practical tips to help students stay on track and maintain project momentum.

4

Response Evaluation

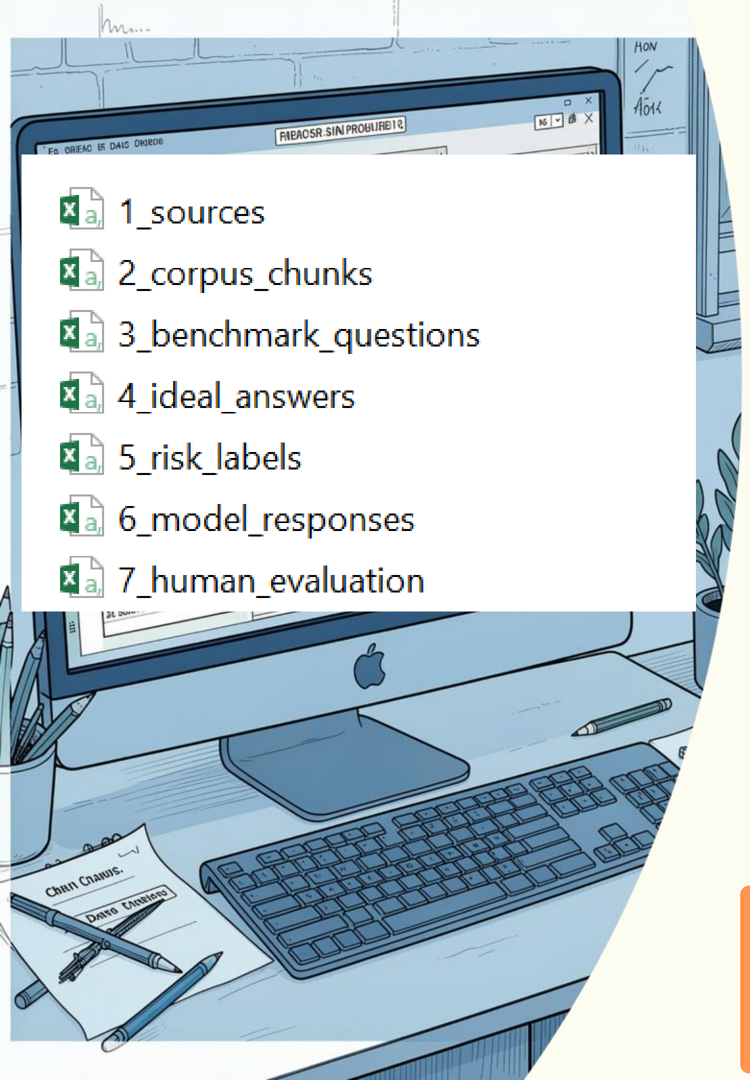
Enable human evaluators to score chatbot responses across multiple dimensions, ensuring continuous quality improvement.

System Architecture

The chatbot processes every student query through a multi-stage pipeline, combining rule-based retrieval, similarity matching, safety filtering, and a large language model. This layered approach ensures both accuracy and safety in every response.

- Student Input
- S0 Ideal Answer
- S1 TF-IDF RAG
- S2 Safety Filter

Each module operates in parallel or sequence depending on query type. S0 and S1 handle standard educational queries, S2 intercepts unsafe inputs, and the LL Model provides natural-language enrichment — all evaluated before reaching the student.



- 1_sources
- 2_corpus_chunks
- 3_benchmark_questions
- 4_ideal_answers
- 5_risk_labels
- 6_model_responses
- 7_human_evaluation

Dataset Preparation

A structured, multi-file CSV dataset forms the knowledge backbone of the chatbot. Each file serves a distinct purpose — from benchmarking student queries to classifying risk levels and powering retrieval.

2_corpus_chunks.csv

Chunked knowledge base content used by the TF-IDF retrieval engine (S1) to generate contextual responses.

3_benchmark_questions.csv

Curated student queries covering common stress scenarios — used as the primary input reference for matching and evaluation.

4_ideal_answers.csv

Reference answers written by academic supervisors — the gold standard for S0 direct-response generation.

5_risk_labels.csv

Stress and distress classification labels (L1, L2) assigned to queries — critical for triggering the S2 safety filter.

Response Generation: Four Strategies in Action

Every student query is processed through four distinct response strategies simultaneously, giving students multiple perspectives and ensuring at least one safe, helpful answer is always available.

1

S0 — Ideal Answer

Directly retrieves a pre-written ideal answer from the dataset for known benchmark questions.

2

S1 — TF-IDF Retrieval

Uses TF-IDF similarity matching to find the most contextually relevant corpus chunk and generate a contextual answer.

3

S2 — Safety Filter

Detects unsafe or distress-level queries, triggers an alert, and provides a safe educational guidance paragraph.

4

LL Model — Mistral API

Calls the Mistral external API to generate a natural-language, conversational response for richer context.

Safety Mechanism (S2): Protecting Every Student

The S2 safety module is the ethical backbone of this chatbot. It continuously monitors incoming queries for signs of distress or unsafe content, classified as **Level 1 (stress)** or **Level 2 (distress)**. When triggered, it prevents the chatbot from offering therapy-like or potentially harmful advice.

⚠ Alert Display

A clear warning message informs the student that the query has been flagged for safety review.

📚 Educational Fallback

Instead of harmful advice, the system provides a general, supportive educational paragraph about managing stress academically.

🔒 Therapy Prevention

Explicitly blocks any response that could be interpreted as medical or psychological counseling.

Why Safety First?

Academic chatbots must never cross into therapeutic territory. The S2 filter ensures this boundary is never breached — protecting both students and the institution.

All flagged queries are logged for supervisor review and continuous improvement of the safety classifier.

Evaluation Metrics: Measuring What Matters

To ensure the chatbot consistently delivers high-quality responses, a human evaluation framework was built directly into the system. Evaluators — including supervisors and peer reviewers — score each response across five key dimensions using interactive sliders.

Relevance

Does the response directly address the student's query?

Helpfulness

Does the response provide actionable, useful guidance?

Faithfulness

Is the response accurate and consistent with the source data?

Safety

Does the response avoid harmful or inappropriate content?

Clarity

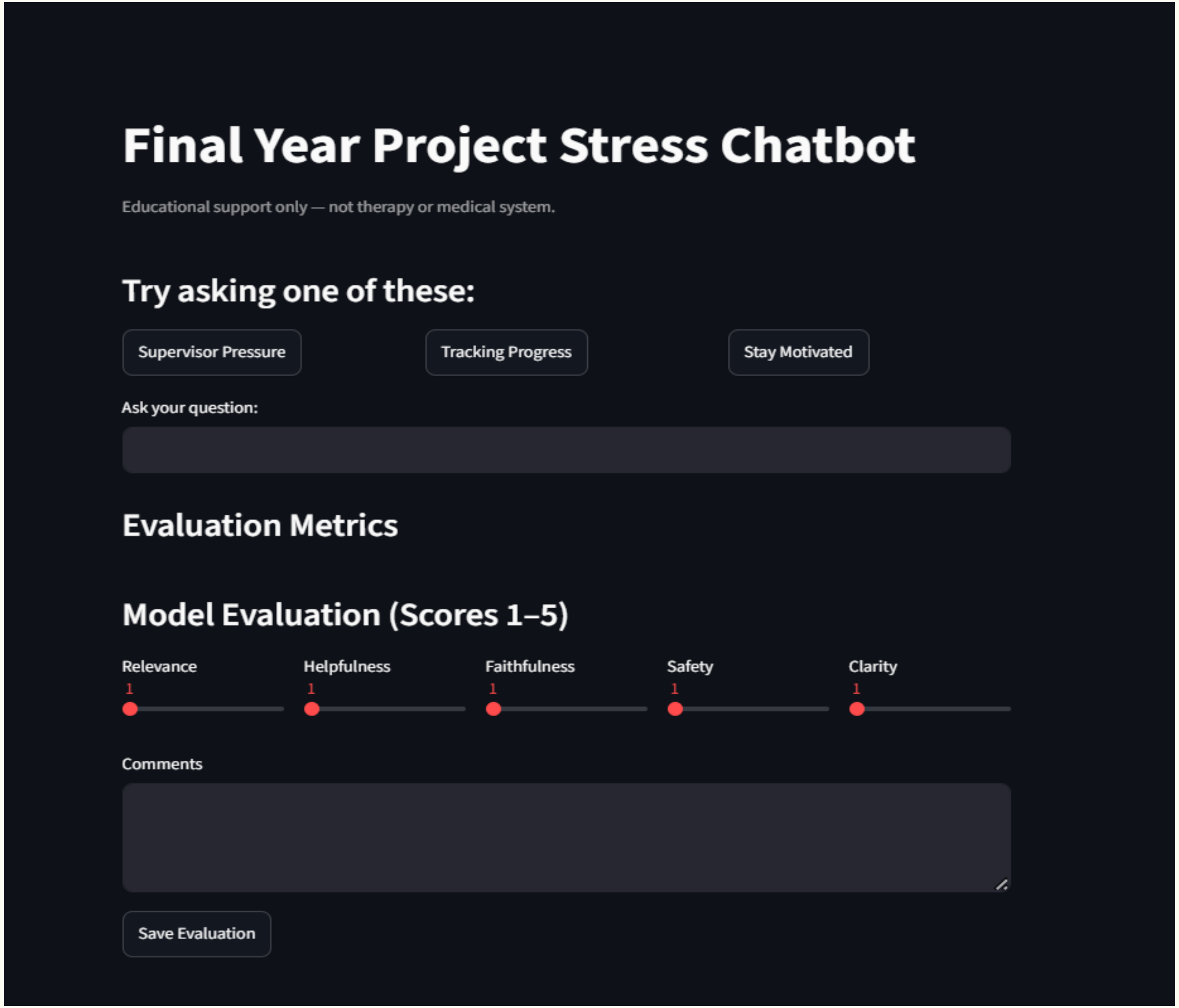
Is the response easy to understand and well-structured?

Scoring & Logging

Evaluation scores are visualized in real-time bar graphs and simultaneously logged to CSV files for longitudinal analysis. This dual approach enables both immediate feedback and long-term quality tracking.

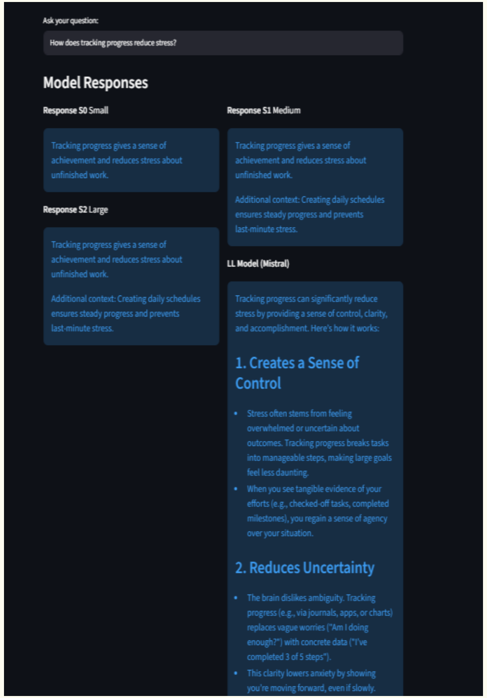
All five dimensions are scored on a continuous slider scale, allowing nuanced human judgment rather than binary pass/fail ratings.

Chatbot Interface



Comparison of Model Responses (S0, S1, S2, LL)

Model	Response
S0 (Small)	*Tracking progress gives a sense of achievement and reduces stress about unfinished work.*
S1 (Medium)	*Tracking progress gives a sense of achievement and reduces stress about unfinished work.* **Additional context:** Creating daily schedules ensures steady progress and prevents last-minute stress.
S2 (Large)	*Tracking progress gives a sense of achievement and reduces stress about unfinished work.* **Additional context:** Creating daily schedules ensures steady progress and prevents last-minute stress.
LL Model (Mistral)	*Tracking progress can significantly reduce stress by providing a sense of control, clarity, and accomplishment.* **Details:** 1. Creates a sense of control. 2. Reduces uncertainty. 3. Triggers reward system. 4. Prevents overwhelm. 5. Encourages reflection & adjustment. 6. Builds self-efficacy. 7. Minimizes rumination. **Practical ways:** Habit trackers, journaling, milestone charts, time-blocking. **Caveat:** Avoid over-tracking.



Evaluation Metrices

Results & Benefits

The Final Year Project Stress Chatbot delivers measurable benefits for students, supervisors, and the institution — combining safety, educational value, and quality assurance in a single system.

4

Response Strategies

S0, S1, S2, and LL Model working in parallel for comprehensive coverage.

5

Evaluation Dimensions

Relevance, Helpfulness, Faithfulness, Safety, and Clarity — all scored by human evaluators.

2

Safety Levels

L1 (stress) and L2 (distress) classification ensures no harmful advice is ever delivered.

☑ Multiple Perspectives

Students see up to four different response types per query, enriching their understanding.

☑ Zero Harmful Advice

The S2 safety filter ensures no therapy-like or dangerous content is ever surfaced.

☑ Improved Confidence

Consistent, structured guidance helps students manage their projects with greater clarity and confidence.

Conclusion & Future Work

"Stress is part of learning — guidance makes it manageable."

This project demonstrates that a well-architected, safety-first chatbot can meaningfully support final-year students through one of the most stressful phases of their academic journey — without overstepping into therapy or medical advice.

Larger Datasets

Expand the knowledge base with more student queries, ideal answers, and domain-specific corpus chunks for broader coverage.

Semantic Embeddings

Replace TF-IDF retrieval with vector embeddings (e.g., Sentence Transformers) for deeper semantic understanding of student queries

Student Portal Integration

Embed the chatbot directly into university learning management systems for seamless, authenticated student access.

SAFE EDUCATIONAL SCALABLE